STATISTICAL INTERPRETATION AND EXPLORATORY DATA ANALYSIS OF THE HUMAN DEVELOPMENT INDEX

CONCEPT AND TECHNOLOGIES OF AI

ASSIGNMENT 1 REPORT

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1. Introduction

The Human Development Index (HDI) is a combined indicator, which was set up by the United nations in order to measure a nation's achievements in three core aspects of human development i.e. Health (calculated through life expectancy at birth), Education (calculated in terms of average and expected years of schooling) and Living conditions (evaluated through Gross National Income per capita). The HDI which measures the development of a country from a much wider perspective then that of the purely economic indicators which makes it more over comprehensive.

This report presents and analyzes the HDI data from 1990 to 2022, with focus on statistical interpretation and exploratory data analysis. The analysis is done with the help of four main objectives given below:

- To evaluate the distribution and statistics of HDI till the latest available year (i.e. 2022)
- To explore short-term trends and variations for period from 2020 to 2022, especially in the light of global events.
- To conduct a detailed analysis, concentrating on South Asian region.
- To map out and compare developmental indicators of the South Asia and the Middle East regions so as to reveal the differences and similarities.

2. Problem 1A – Single Year HDI Exploration (2022)

At first, a clone of the database was made and then the clone database underwent filtering in order to eliminate the records of years and only keep the data from 2022. During the processing cycle, the missing values were filled, the duplicate data were removed, and the numeric columns were placed in right format for upcoming calculations. After managing the data, the descriptive statistics were calculated to get Mean, Median, Standard deviation, and the country with the highest and lowest HDI.

Key Results

The important metrics obtained from the 2022 statistical analysis are:

- Mean HDI: Approximately 0.73,
- Median HDI: Approximately 0.74,
- Standard Deviation: Approximately 0.07,
- Country with the highest HDI is Norway (about 0.96), and
- Country with the lowest HDI is Niger (about 0.45).

Also, those countries that scored more than 0.800 on the HDI were put in the "Very High Human Development" category.

The downward compilation of countries based on the Gross National Income per capita in the "Very High" HDI rating category table, which topped the list of ten countries was one of the analytical procedures. Additionally, a list was created to show the frequencies of the countries in the distribution of the various HDI categories (i.e. Low, Medium, High and Very High).

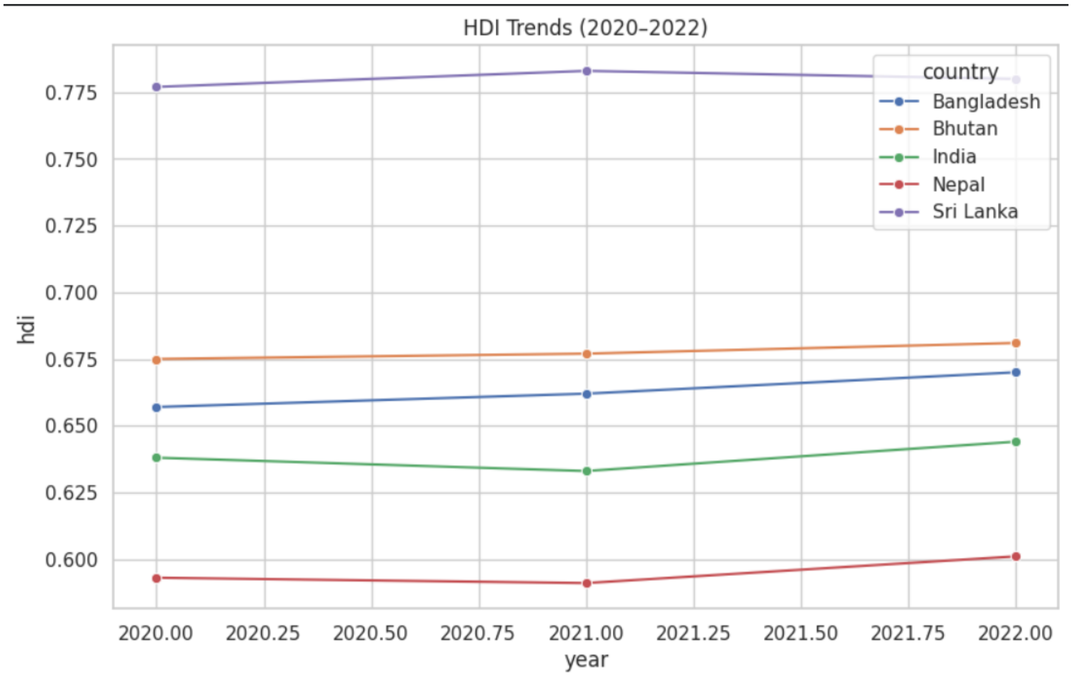
Therefore, it can be interpreted from the data that nations with strong education and health care resources can still score high in HDI even if their incomes are not so high. This supports the argument that economic wealth is not the only factor determining high human development, and social investment are equally important.

3. Problem 1B – HDI Trend Analysis (2020 – 2022)

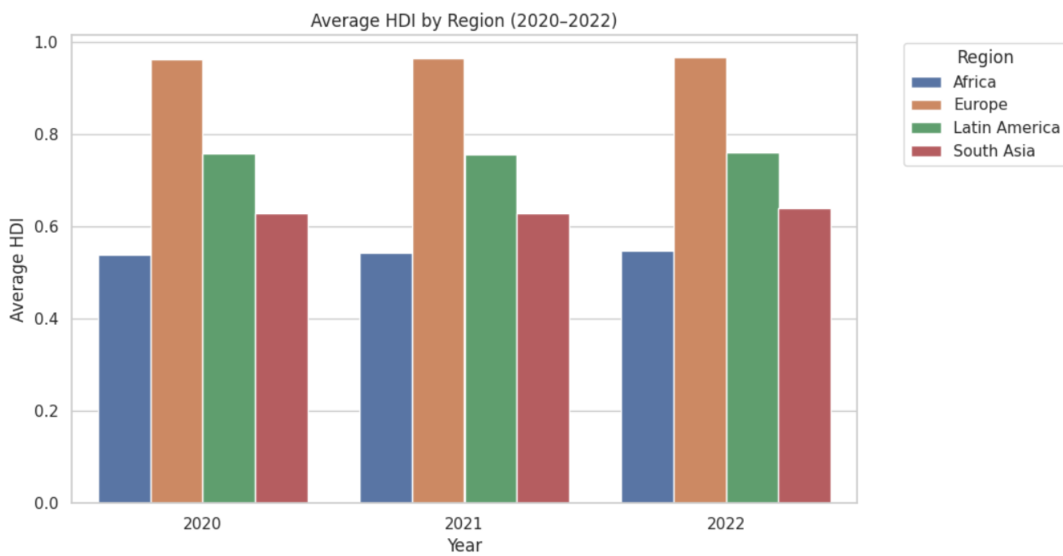
Again, another clone of the database was made and then the clone database underwent filtering in order to collect data from only three years 2020-2022. The filtering process included fixing the missing values and standardizing the country names for uniformity throughout the years. Then, the trends and correlation for five random countries were observed by using four different types of visualizations, i.e. line chart, bar chart, box plot and scatter plots.

Key Results

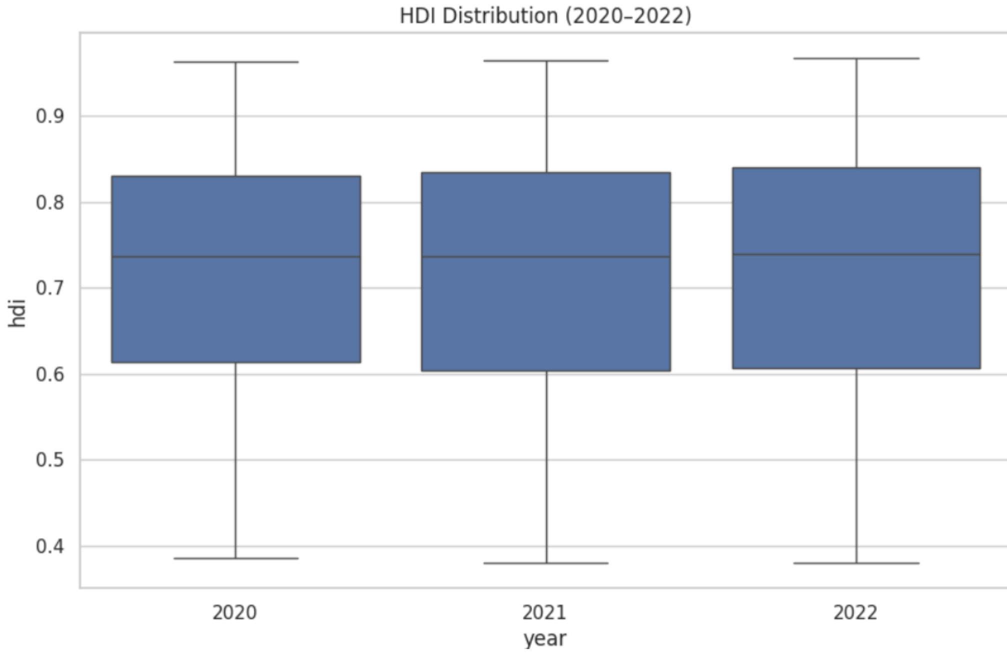
- Line Chart: Nepal and Bangladesh continued to improve steadily, while Sri Lanka remained at a standstill.



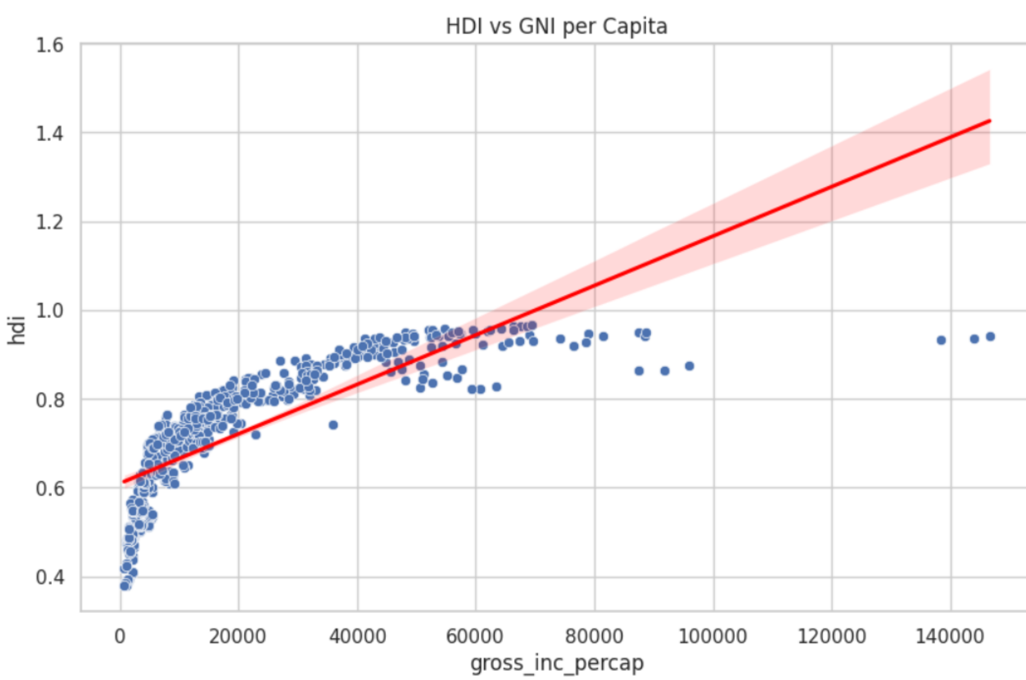
- Bar chart: At the regional level, Europe was found to have the highest HDI average, while the lowest was assigned to Sub-Saharan Africa.



- Box plot: The world's median HDI experienced a very small increase from 2020 to 2022, thereby signifying the recovery trend.



- Scatter plot: The strong positive correlation between GNI per capita and HDI scores was clearly defined.



Therefore, the analysis shows that the COVID-19 pandemic caused a disruption of the expected HDI growth during 2020-2021. However, the 2022 figures do already show recovery signs. It is particularly the case of the richer areas that have overtaken the poorer areas regarding the post pandemic recovery, pointing to an uneven situation.

4. Problem 2 – Advanced HDI Exploration (South Asia)

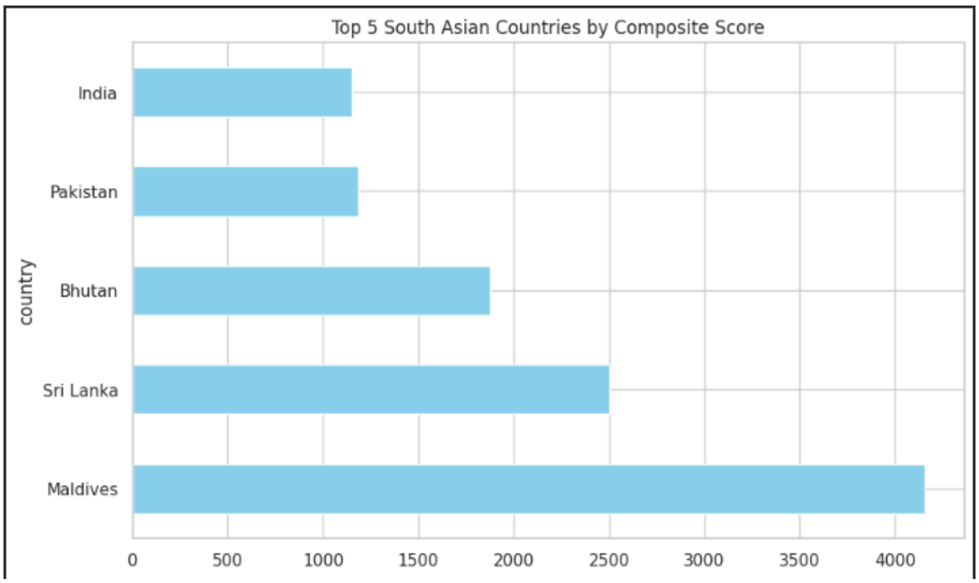
Eight South Asian Countries were selected and a subset of them isolated to study in detail. A composite score of its own was computed by means of the formula:

0.30 x Life Expectancy + 0.30 x GNI Index.

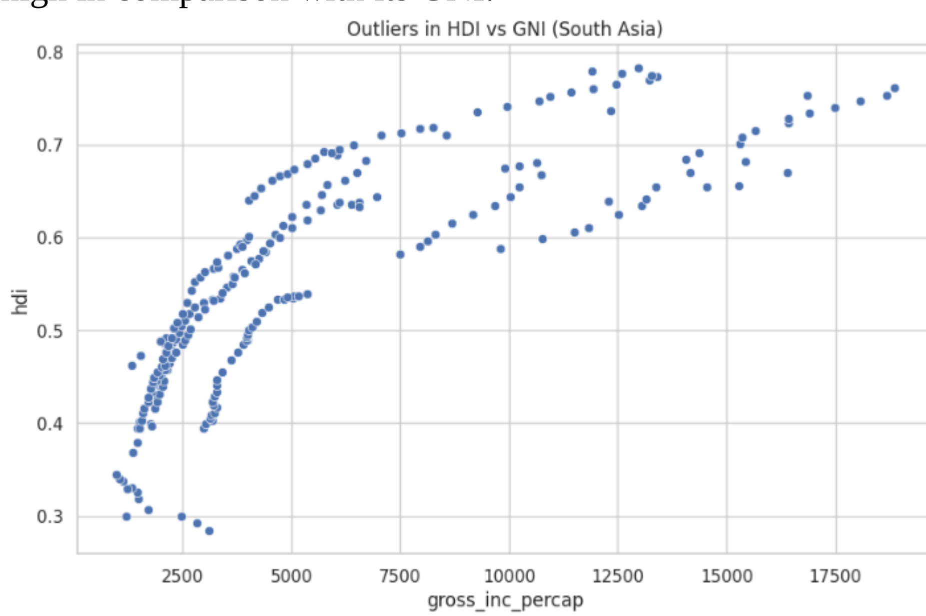
The 1.5 x IQR (Interquartile Range) rule was used to identify the outliers. It was correlated with Gender Development and Life Expectancy variables and a Gap Analysis (GNI rank less HDI rank).

Key Results

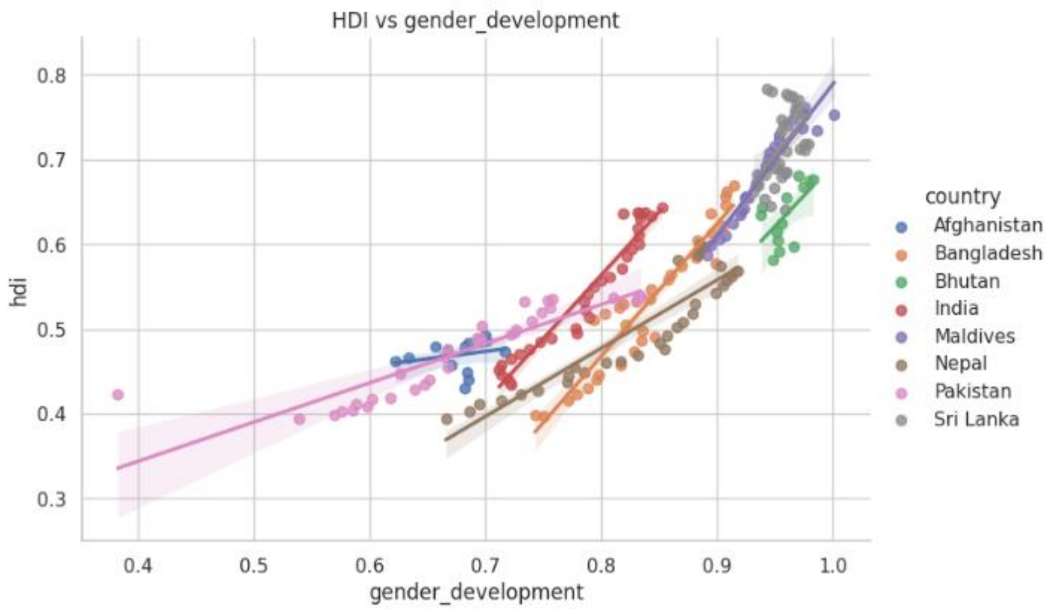
- Composite Score Ranking: the highest score was awarded to Maldives and Sri Lanka, whereas the lowest score was awarded to Afghanistan.

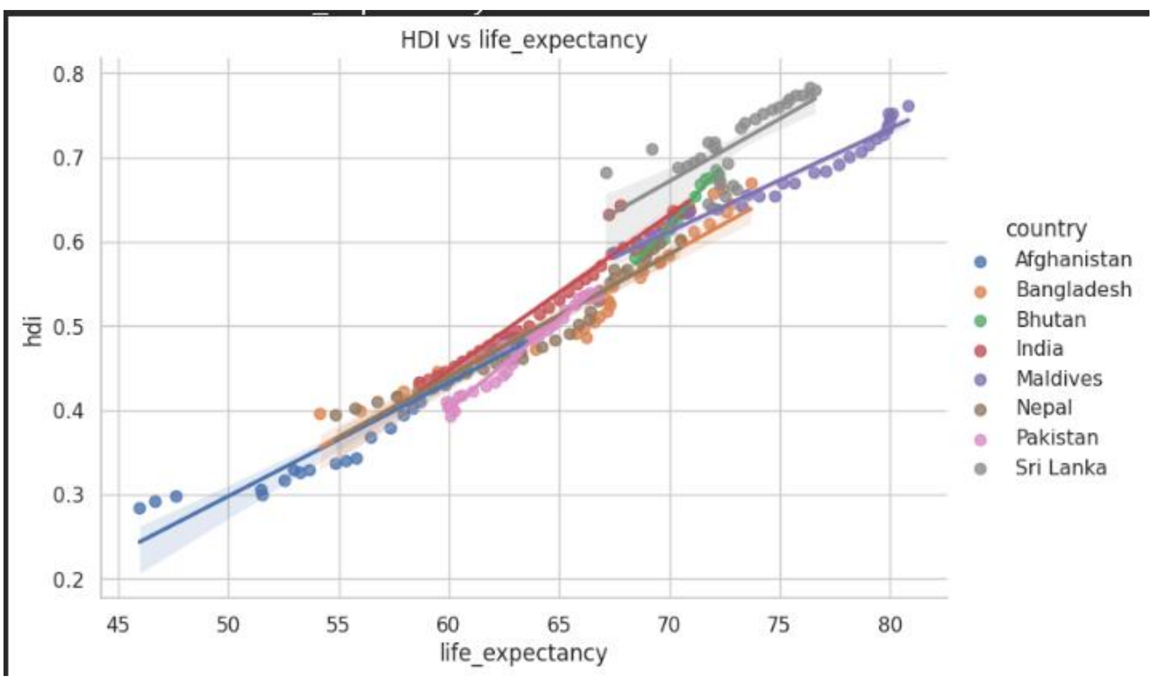


- Outliers: The Maldives was declared to be an outlier because its HDI was unnaturally high in comparison with its GNI.

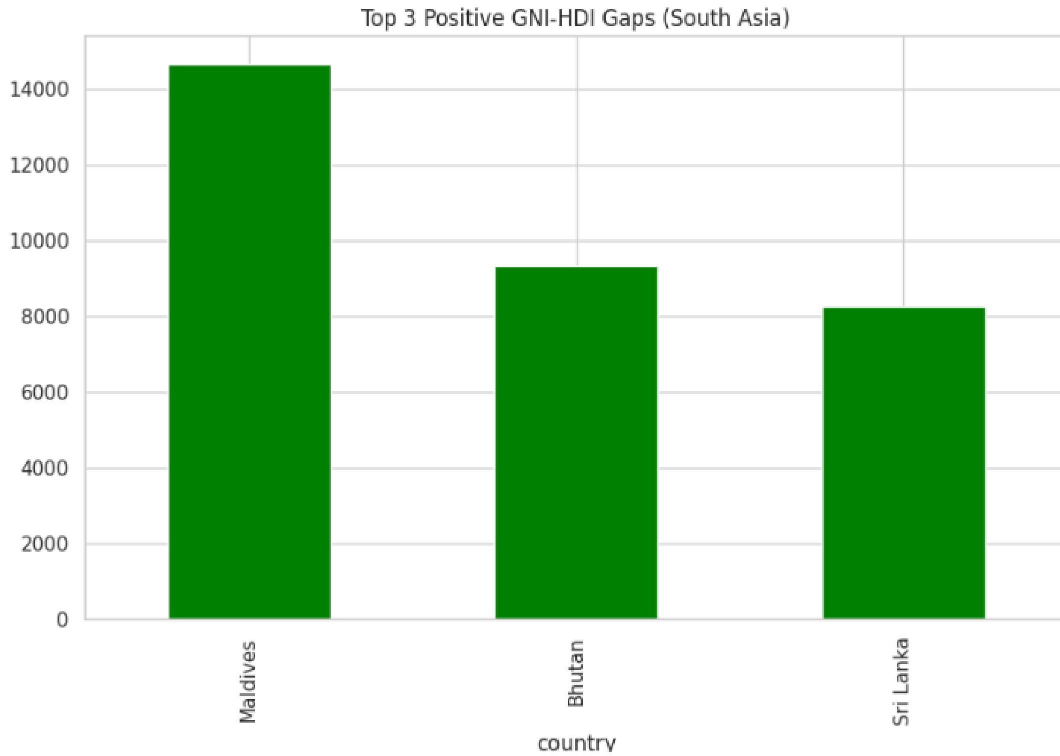


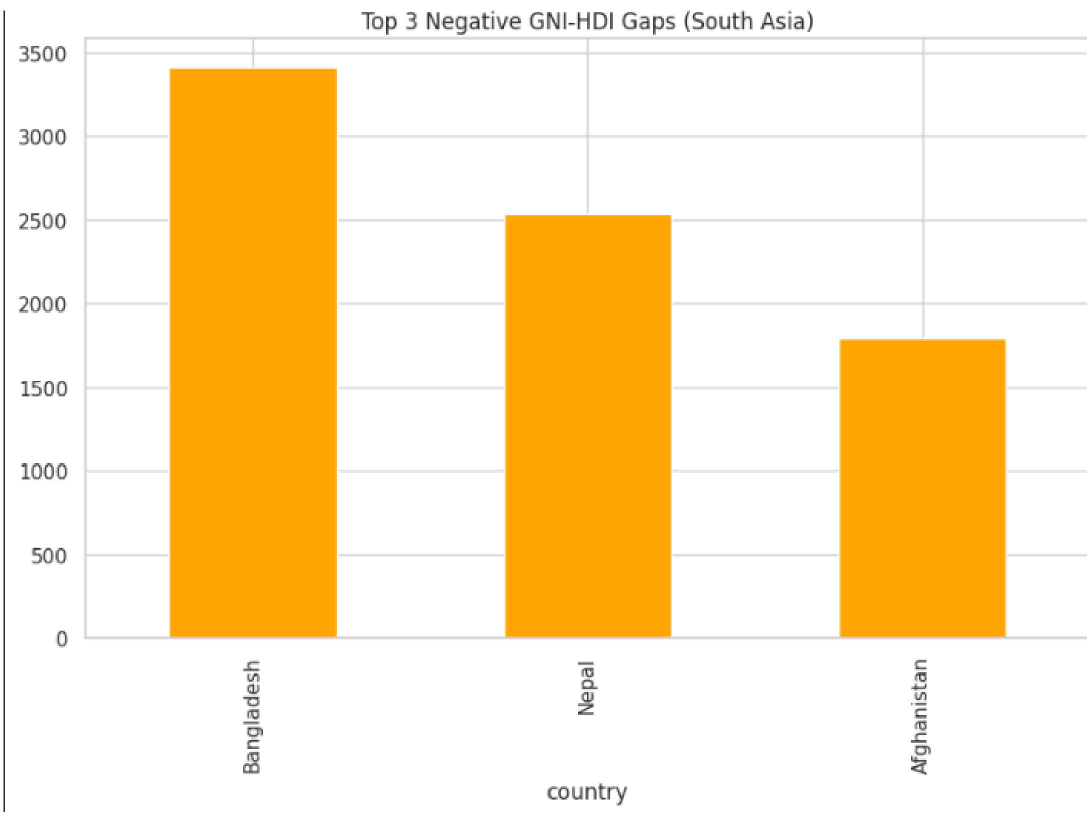
- Correlations: There was a strong correlation between Life Expectancy and HDI (r approximately 0.85), but a weaker correlation between Gender Development and Life Expectancy (r approximately 0.55).





- Gap Analysis: Gap in India was positive (income ranking above HDI ranking), whereas in Nepal was negative (HDI ranking above expected based on income).





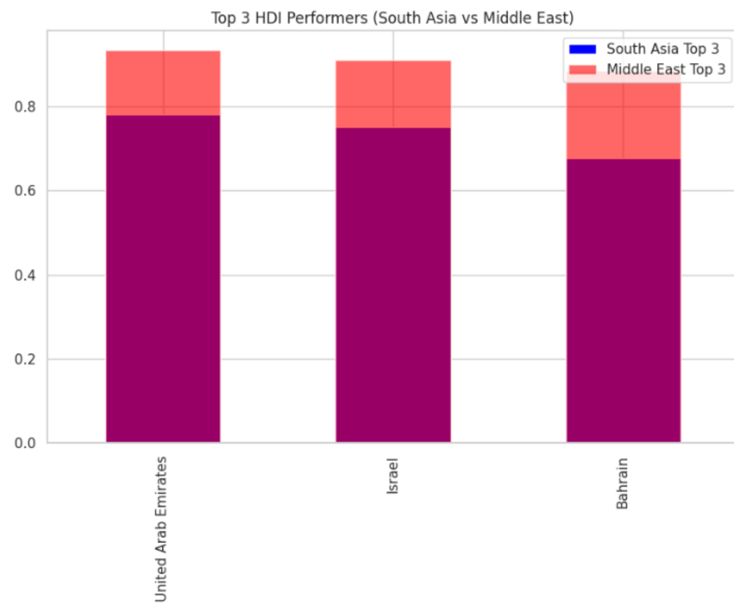
South Asia portrays diversity in its development trends. The results highlight the importance of the fact that economic development does not necessarily imply human development. Such nations as Nepal show that investments in such non-economic sectors as health and education may increase HDI, even in countries with lower income levels.

5. Problem 3 – Comparative Regional Analysis (South Asia vs Middle East)

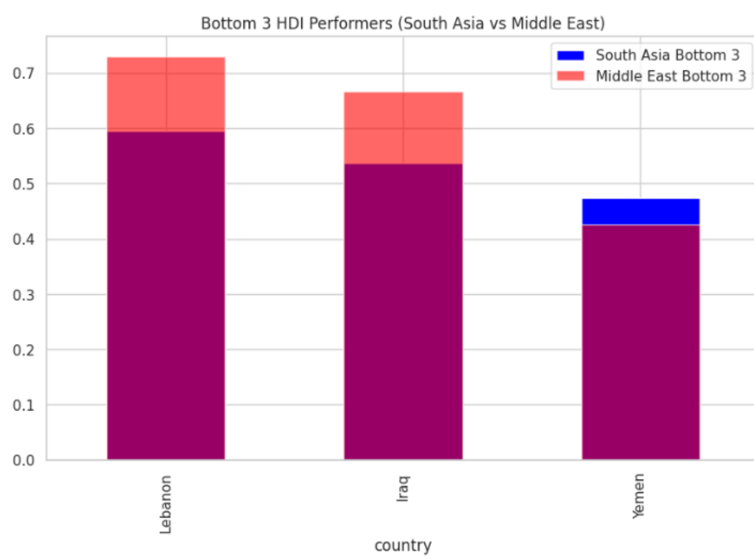
They were broken down into two different subsets; South Asia (8 countries) and Middle East (14 countries). This analysis entailed the calculation of descriptive statistics, the best and the worst performers, comparisons of important metrics, calculation of disparities, and determination of correlations and outliers in each region.

Key Results

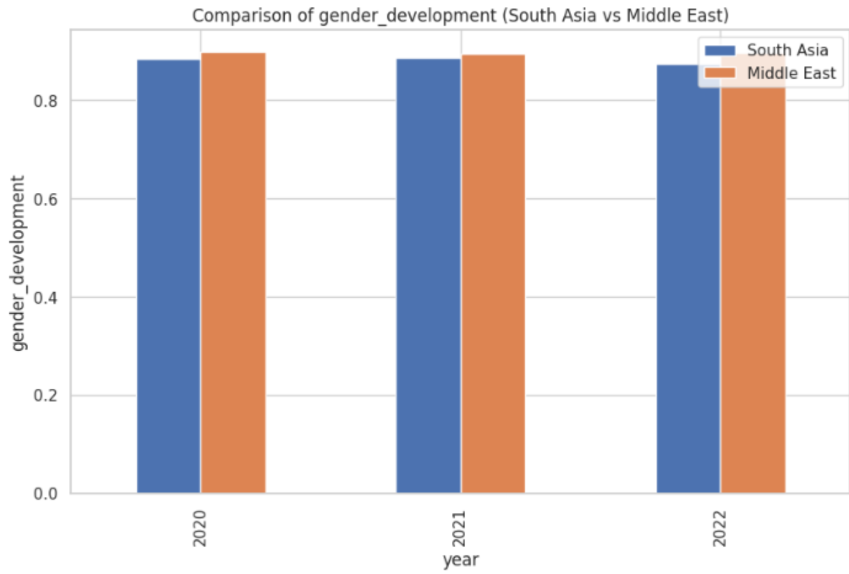
- Means: The Middle East value of HDI (approximately 0.80) is much greater than the South Asia (approximately 0.72).
- Best in Show: Qatar, UAE, and Israel are the best in the Middle East; Maldives and Sri Lanka are the best in South Asia.

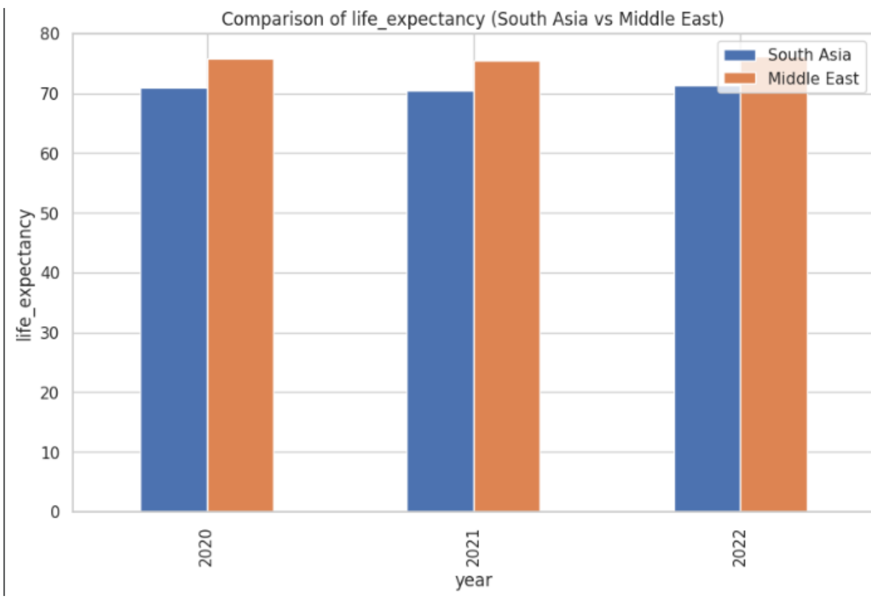


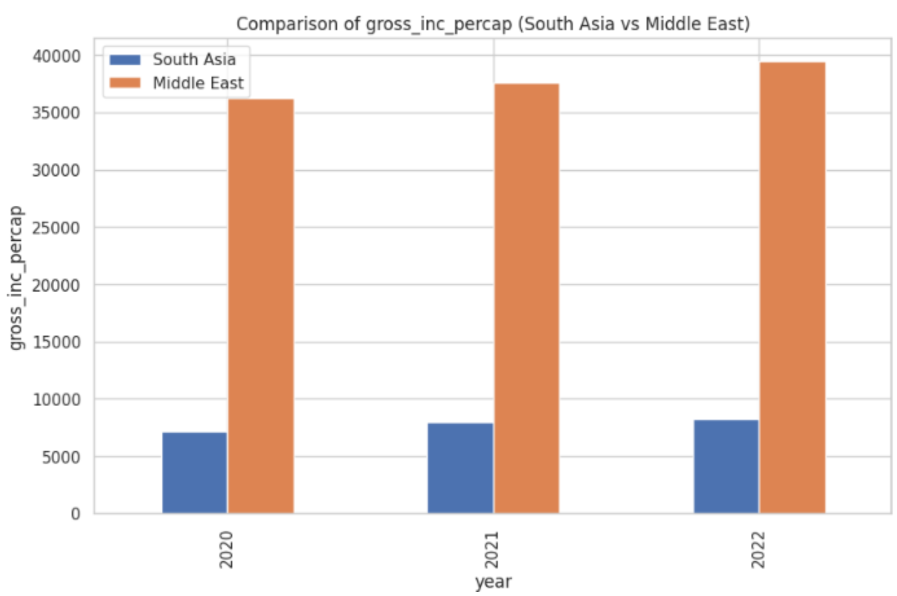
- Impoverished Achievers: Yemen and Syria (Middle East) against Afghanistan and Pakistan (South Asia).



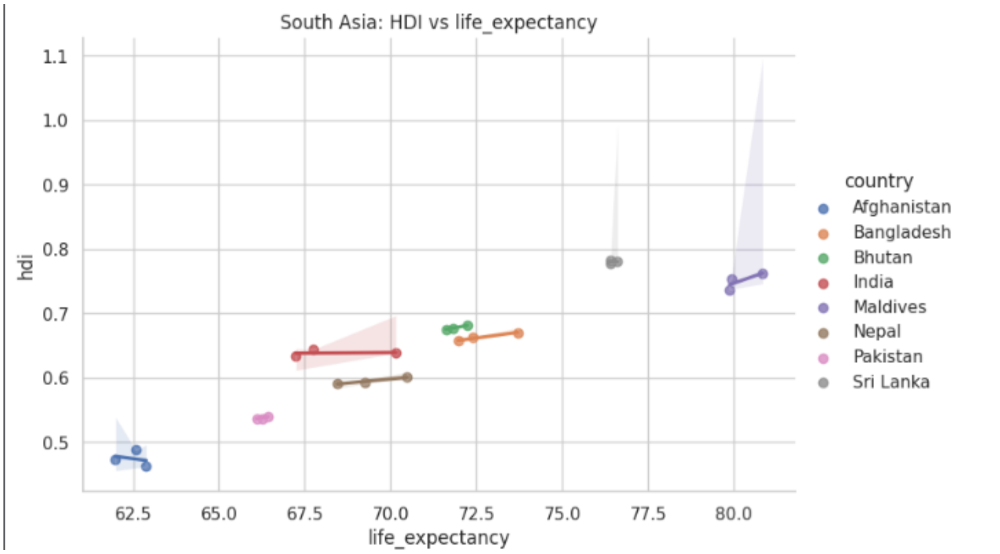
- Metric Comparison: The variable of GNI per capita is the most disparate variable of the two regions.

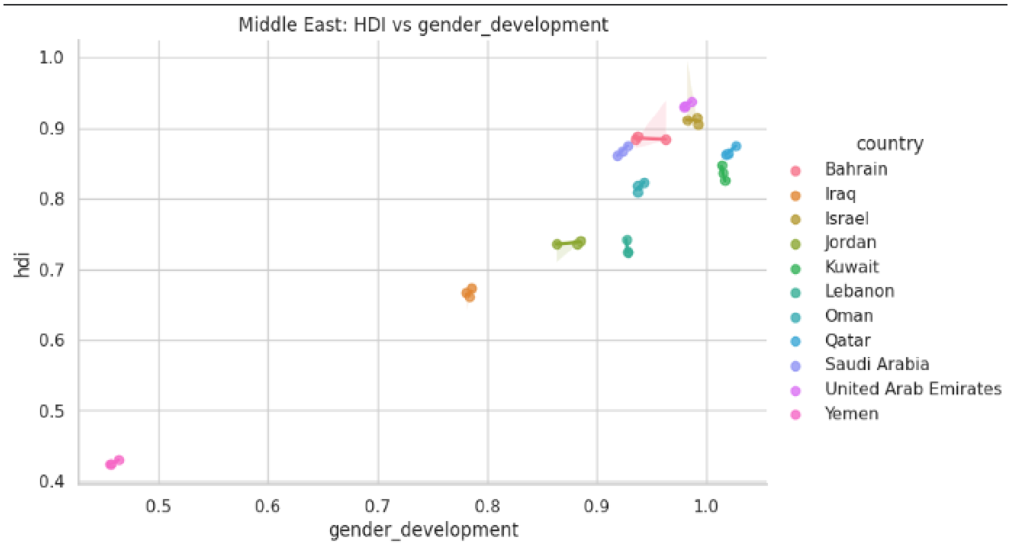


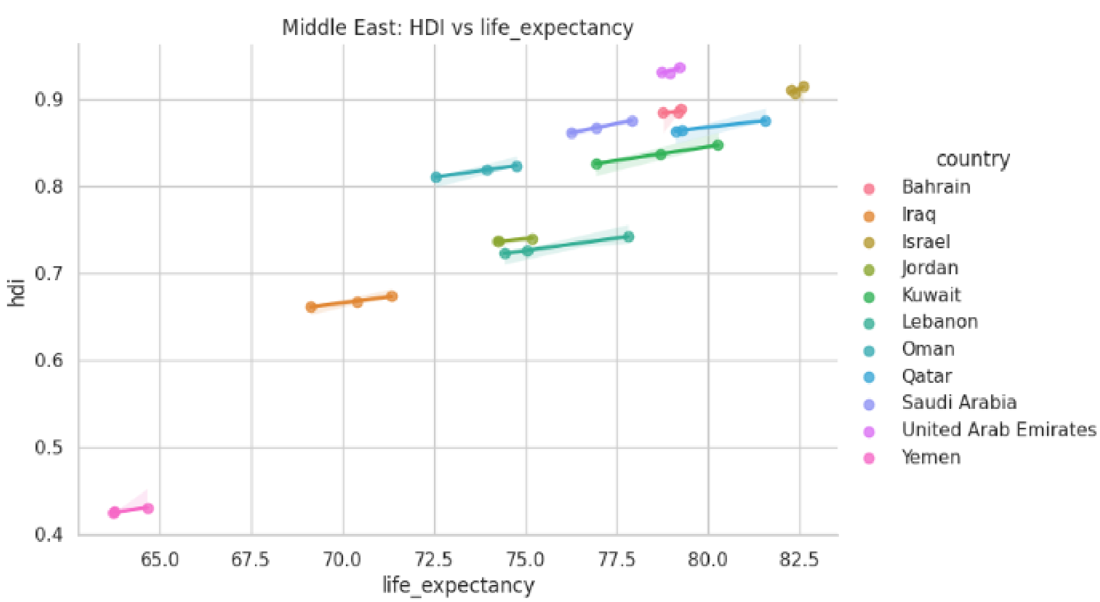




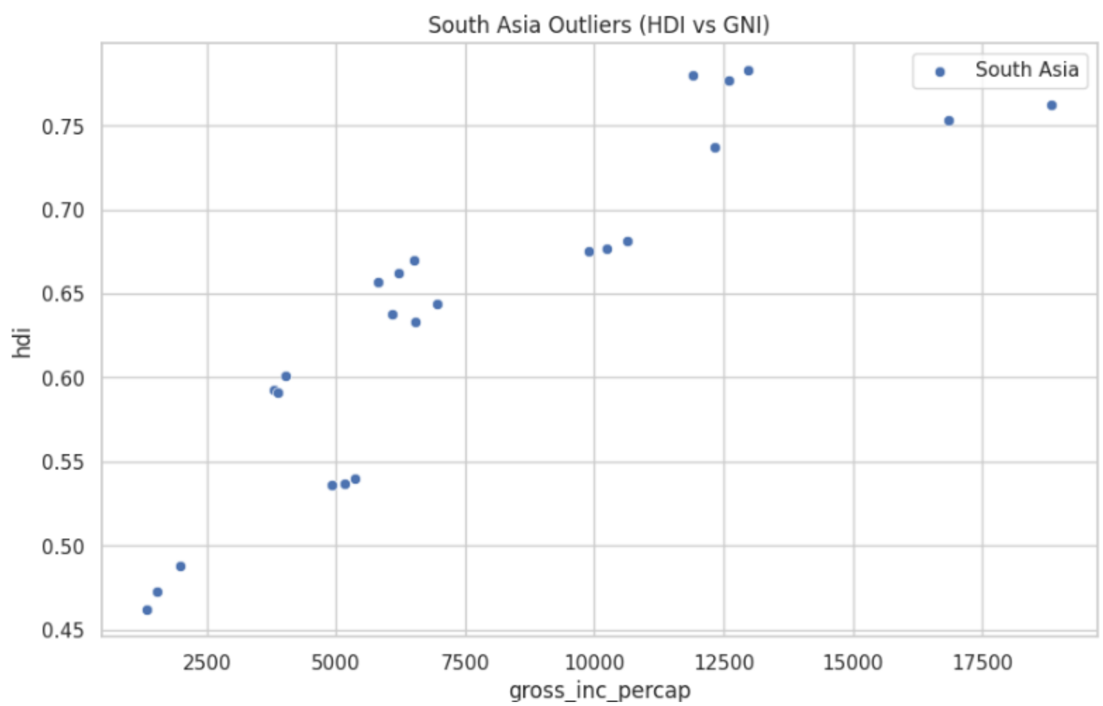
- Correlation: South Asia exhibits greater internal variation (CV approximately 0.10) than in the Middle East (CV approximately 0.08) which implies a more skewed development in South Asia.

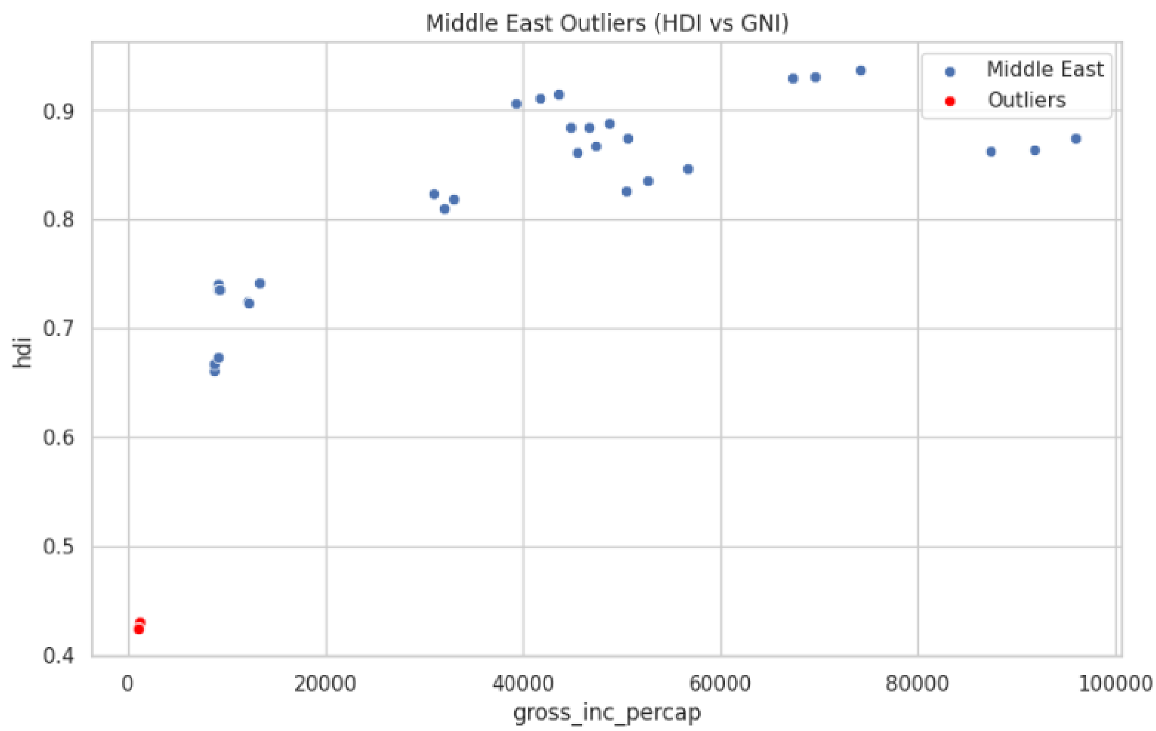






- Outliers: Yemen and Afghanistan are negative outliers, which can be explained by conflict and instability to a large extent.





The benefits of natural resources in the Middle East are leading to an increase in the average level of HDI, but conflict regions are considerably lowering the performance of the region. HDI growth is more consistent but slower in South Asia, as compared to the Middle East, where it is mainly caused by economic factors.

6. Conclusion

In conclusion, The HDI scores differ significantly in different countries and regions through different degrees of investment in health and education. Although income is a powerful determinant of HDI, health and education indicators account to the high disparities in the nations pursuing similar GNI levels. COVID-19 made the world slow down HDI growth before a disproportionate recovery in favor of richer countries. Most of South Asia is typically underdeveloped when compared to Middle East, even though, a particular country in the region is doing better than the expected by its income.

7. Appendix

The following appendix gives a brief summary of all the data that was examined in this report.

2022 HDI Overview:

The world average HDI in 2022 is 0.73 which has a median of 0.74, indicating that the majority of the countries are concentrated in middle of human development. Norway topped the globe with HDI of 0.96 which is excellent in health, education, and income. On the other hand, Niger has the lowest HDI of 0.45, which shows that developmental issues are a major problem. Human Development nations that are considered to be very high (HDI is a higher number than 0.80) fall under this category and are usually the ones that have good health care systems and quality educational infrastructure, despite that fact that their income are in the middle range.

HDI Trends (2020-2022):

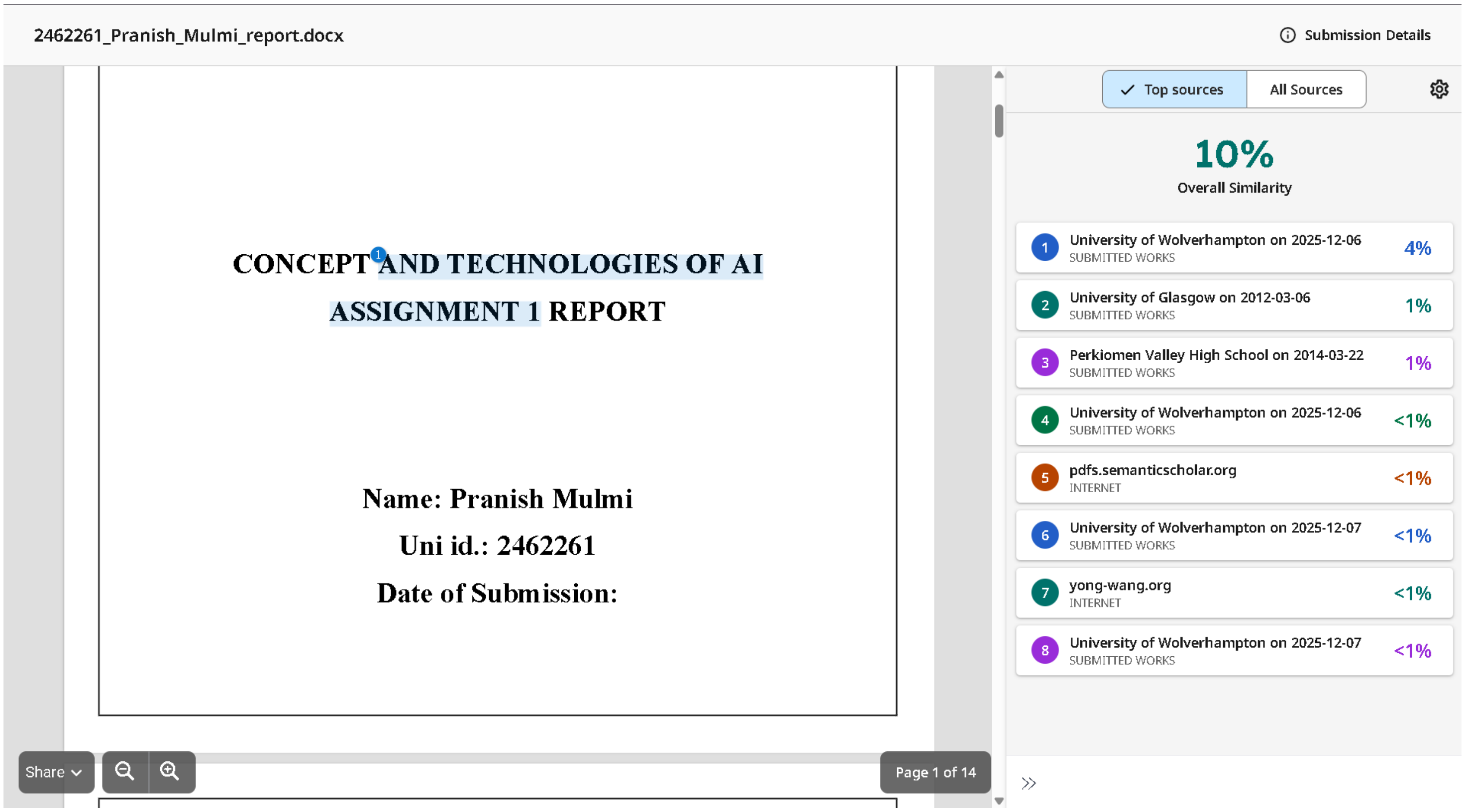
The three years 2020 to 2022, provided visibility of how COVID-19 affected the development of the world. In the year 2020-2021 mist countries in the South Asian regions such as Nepal and Bangladesh steadily improved even during the pandemic, whereas the HDI of Sir Lanka was completely stable. Europe had the highest regional average of HDI whereas Sub-Saharan Africa remained to have the most developmental problems. GNI per capita and HDI had a very strong positive correlation, which validated the fact that richer countries tend to have more favorable HDI results.

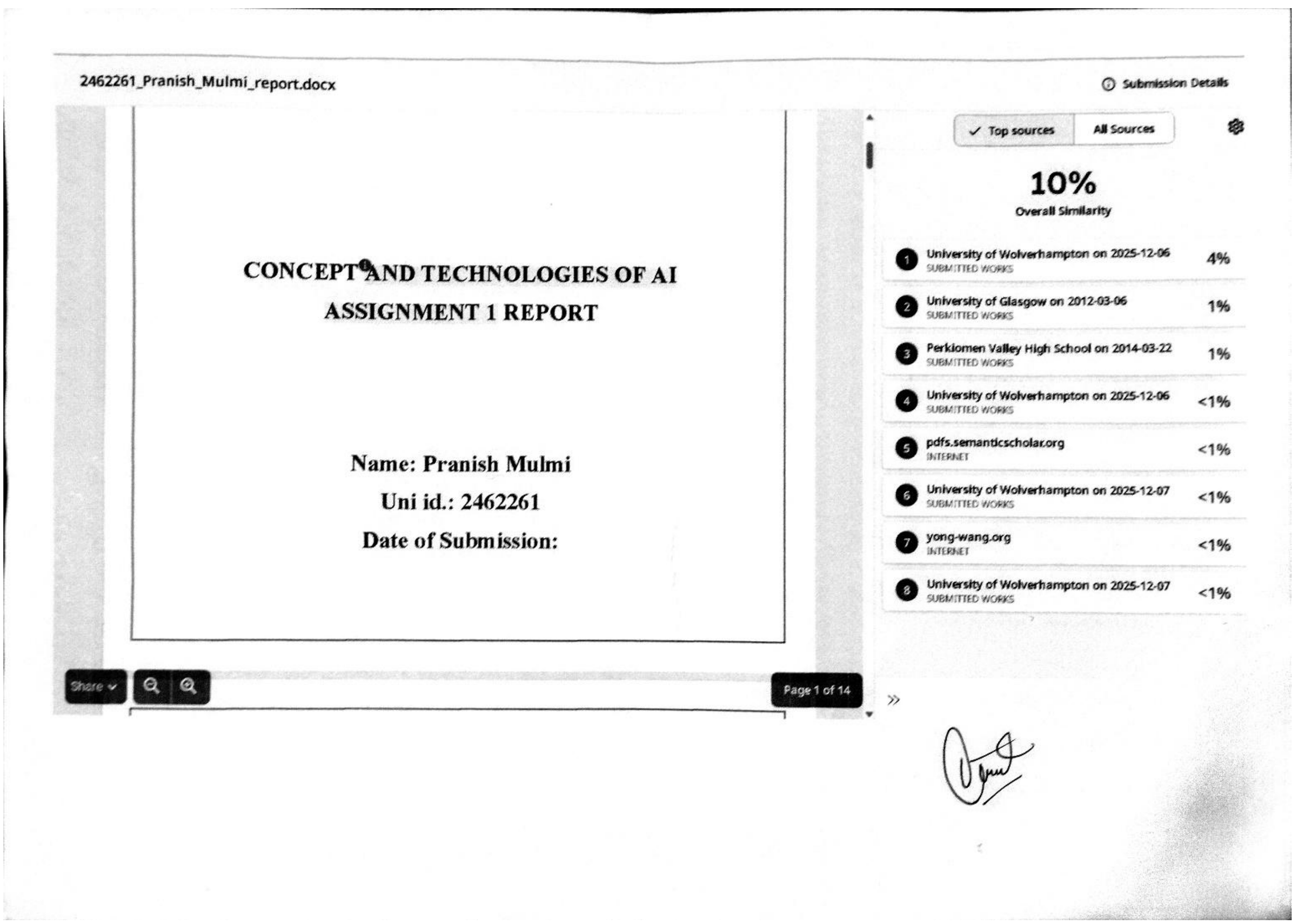
South Asia Analysis:

South Asia is a varied developmental scenario with eight countries performing different developments. Maldives and Sri Lanka are the countries with the greatest composite scores of life expectancy and an income measure of the region. Afghanistan is the least developed due to constant war and turmoil. The fascinating fact is that the Maldives is more of an outlier with HDI much greater that one would think as a result of GNI alone implying that the country is making good investments in health and education. In this region, life expectancy is considerably correlated with HDI (correlation coefficient 0.85), whereas the same cannot be said of gender development (correlation coefficient 0.55).

Comparison (South Asia vs Middle East):

South Asia is doing much poorly than the Middle East with its average HDI at 0.80 as compared to 0.72. Maldives and Sri Lanka rank first in South Asia, where as Gulf countries such as Qatar, UAE, and Israel rank first in the Middle East. Nonetheless, failed countries with severe problems both belong to the Middle East: Yemen and Syria, and South Asia where the political instability and war are the primary reasons. The biggest gap between the two regions is the GNI per capita that shows the oil abundance of the Middle East. It is also interesting to observe that South Asia has a higher coefficient of variance (i.e. 0.10) as compared to the Middle East (i.e. 0.08) meaning that there is more in South Asia.





GIT – LINK: https://github.com/Pranish-Mulmi/concepts-and-technologies-of-ai